

Name of the Examination: WINTER 2022-2023 – CAT-2

School : School of Electronics Engineering
Course Code : CSE2013 Course Title : Information Theory and Coding
Set Number : 02 Date of exam : 28/03/2023 (FN)
Duration : 90 Min Total Marks : 50
(B1)

Answer all the questions

Marks

- Q1** Consider a discrete memoryless source $S = [\alpha, \beta, \gamma, \delta, \theta, \eta, \mu]$ and their respective probabilities as $[0.3, 0.46, 0.06, 0.03, 0.12, 0.01, 0.02]$. Using Huffman code find the coding efficiency and redundancy. **12**
- Q2** Consider the source symbols $S = \{A, B, C\}$ and their respectively probabilities are $P(A) = 0.7, P(B) = 0.2$ and $P(C) = 0.1$. Using Shannon-Fano coding, compare the coding efficiencies of S and S^2 . Also using the codewords of S^2 decode the binary data 110010100010011110 in terms of X_i : (Consider X_1 to X_9 in decreasing order of the probability of S^2). **14**
- Q3** A DMS outputs the symbols from the source alphabet $\mathcal{X} = \{x_1, x_2, x_3, x_4, x_{end}\}$ with probabilities $\{0.15, 0.25, 0.2, 0.3, 0.1\}$ respectively. Here x_{end} is the end-of-symbol. The data has been encoded using arithmetic coding. The received tag value is 0.5429. Find the decoded data. **12**
- Q4** Decode the non-binary codeword "75 77 79 81 257 258 257 81 259 258 81" into a 4×4 image patch using LZW algorithm. (Also construct the dictionary by considering initial dictionary as 0-255). **12**

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	PO2	PEO1	PSO2	12
Q2	3	CO3	PO4	PEO1	PSO1	14
Q3	3	CO3	PO4	PEO1	PSO3	12
Q4	3	CO3	PO4	PEO1	PSO1	12

Name of the Examination: WINTER 2022-2023 – CAT-2

School : School of Electronics Engineering
Course Code : CSE2013 Course Title : Information Theory and Coding
Set Number : 04 Date of exam : 28/03/2023 (An)
Duration : 90 Min Total Marks : 50 (B2)

Instructions: Assume data wherever necessary. Any assumptions made should be clearly stated.

Marks

- Q1** Apply Huffman coding for a zero memory source emitting the symbols $\{S_i\}=\{S_1, S_2, S_3, S_4, S_5, S_6, S_7\}$ with probabilities as shown in table, Encode the data for the sequence of data transmitted as “S₃, S₂, S₇, S₂, S₄, S₂, S₁, S₅”. **12**

S_i	S_1	S_2	S_3	S_4	S_5	S_6	S_7
$P(S_i)$	0.05	0.13	0.08	0.17	0.30	0.02	0.25

- Q2** Compute the tag value of the given word “Happy” using Arithmetic coding in a binary form. **14**

- Q3** A message has 8 symbols namely $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$ with probabilities as 0.5, 0.125, 0.125, 0.0625, 0.0625, 0.0625, 0.03125 and 0.03125 respectively. Apply Shannon Fano coding and calculate the average code length, efficiency and redundancy. **12**

- Q4** Decode this code 5,2,4,1,8,6,9,10,3,11,12,2,14 and 3 using LZW dictionary decoding. If the initial dictionary as **12**

Entry	Index
1	@
2	&
3	*
4	#
5	!

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	2, 4	1	1	12
Q2	3	3	2, 4	1	1	14
Q3	3	3	2, 4	1	1	12
Q4	3	3	2, 4	1	1	12

Name of the Examination: WINTER 2022-2023 – CAT-2

School : School of Electronics Engineering
Course Code : CSE2013 Course Title : Information Theory and Coding
Set Number : 05 Date of exam : 27/03/2023 (AN) (A2)
Duration : 90 Min Total Marks : 50

Instructions: Assume data wherever necessary. Any assumptions made should be clearly stated.

Marks

- Q1** Consider a discrete memory source with seven possible symbols x_i , $i=1, 2, 3, \dots, 7$, and the corresponding probabilities as 0.46, 0.06, 0.30, 0.03, 0.02, 0.01 and 0.12. Construct the Huffman code and calculate the coding efficiency and redundancy. **12**
- Q2** If the initial dictionary is given as shown in the table. Encode the data transmitted using Lempel-Ziv-Welch coding technique. The transmitted message is "ZQRRQS ZQRRQS ZQRRQS ZT". **12**

Entry	Index
1	S
2	Q
3	R
4	T
5	Z

- Q3** Apply Shannon Fano coding for following message and calculate the code length, efficiency and redundancy. A message has 8 symbols $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}$ has probability 0.03125, 0.12, 0.13, 0.25, 0.0625, 0.0625, 0.125, 0.125, 0.0625 and 0.03125 respectively. **14**
- Q4** For given set of symbols {'e', 's', 't', 'b', '.'} the probabilities are {0.3, 0.2, 0.1, 0.3, 0.1} respectively, compute the word for the given tag value 0.81611 using Arithmetic decoding. **12**

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	2, 4	1	1	12
Q2	3	3	2, 4	1	1	12
Q3	3	3	2, 4	1	1	14
Q4	3	3	2, 4	1	1	12

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Set Number : 06 Date of exam : 27/03/2023 (FN)
Duration : 90 Min Total Marks : 50

(A1)

Instructions: Assume data wherever necessary. Any assumptions made should be clearly stated. **Marks**

Q1 A Source produce two symbols with probabilities $3/4$ and $1/4$. Design a variable length code using Shannon-Fano algorithm, so that the coding efficiency is at least 85% **14**

Q2 Consider a sequence of letter from English alphabet with their probability of occurrence as **12**

A I X M N O P Y
0.1 0.1 0.2 0.1 0.1 0.2 0.1 0.1

(a). Construct binary Huffman code by moving the combined symbol as high as possible. Also find efficiency, average code word length and variance of coding.

(b). Construct binary Huffman code by moving the combined symbol as low as possible. Also find efficiency, average code word length and variance of coding.

Q3 Decode this code “3, 1, 4, 6, 8, 4, 2, 1, 2, 5, 10, 6, 11, 13, 6” using LZW dictionary, with initial dictionary given below **12**

1	h
2	o
3	p
4	e

Q4 Encode the message ‘NEWS’ using an arithmetic coding algorithm with the following probabilities of occurrence of each symbol. **12**

Symbol: E W S N
Probability of occurrence: 0.3 0.3 0.2 0.2

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	1	1	2	1	14
Q2	3	2	2	2	1	12
Q3	3	1	1	2	1	12
Q4	3	2	4	-	1	12